



ESCUELA DE INGENIERÍA
FACULTAD DE INGENIERÍA

Lecture 4: Consumer Theory II

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ICS2523 Microeconomics

Lecture 4

Last lecture we

- Compared Partial Equilibrium Models vs. General Equilibrium Models
- Started with **Consumer Theory**
 - Preferences and Utility
 - Indifference Curves
 - Define the Marginal Rate of Substitution (MRS)

In today's lecture, we will

- Learn about the Budget constraint
- Understand consumer's choice
- Learn a **mathematical approach** to model consumer's choice

Questions from last lectures

Lecture 2

- Ceteris Paribus: we assume that everything outside of the problem at hand remains constant. Regarding this, I actually do wonder how to know where to draw the line (which variables can be considered as "inside" the problem?)

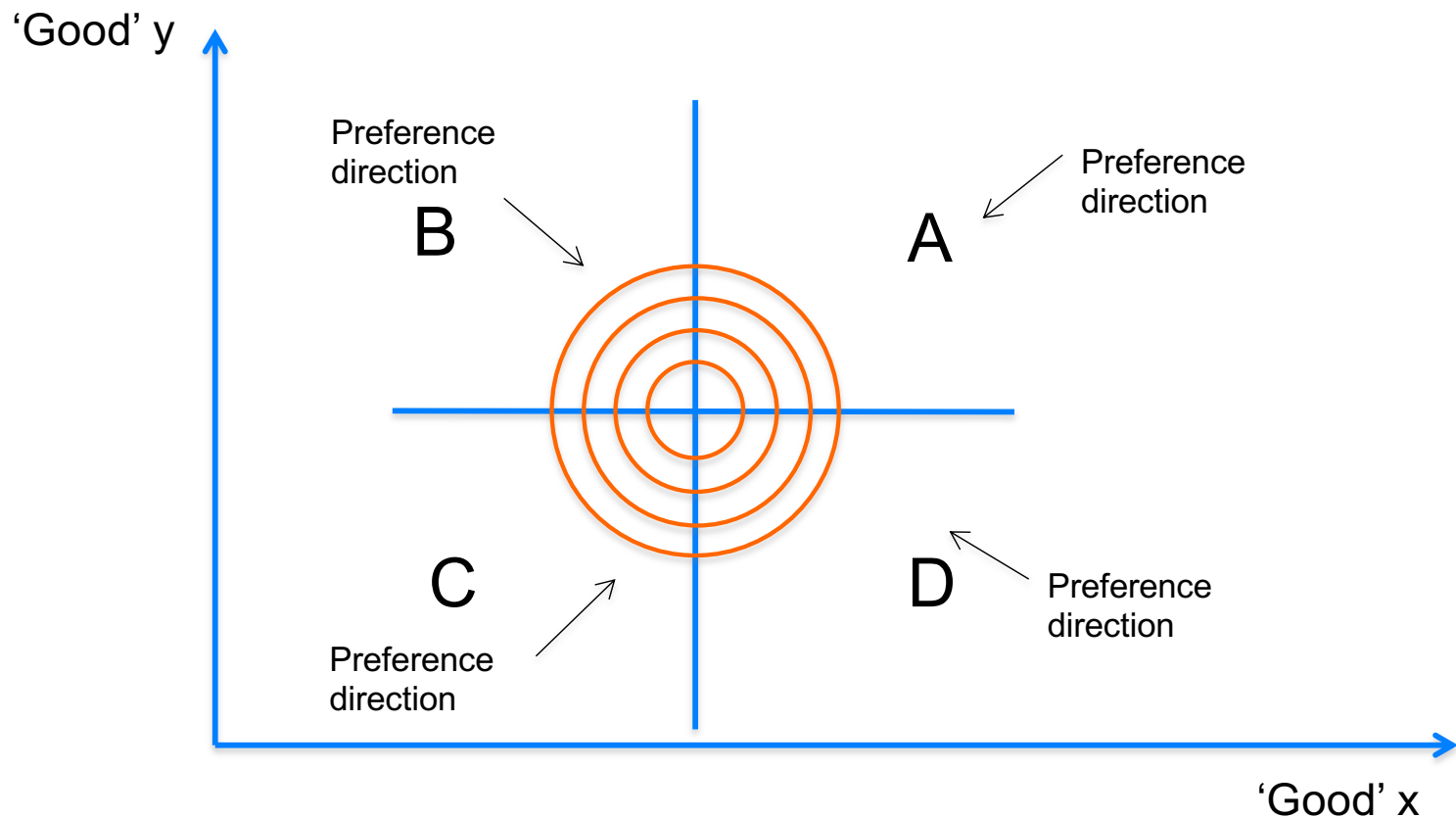
Lecture 3

- Example of commodities
- Explain again slides of goods/bads

Good, bads, and neuters

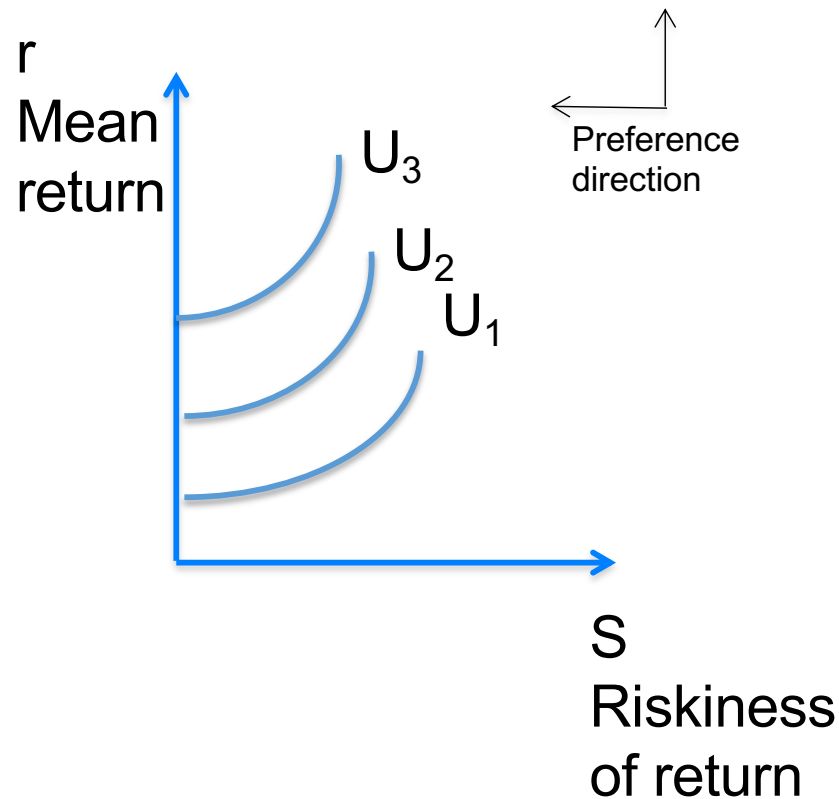
Not all commodities are goods (pollution is a bad, for example)

Also, everyday experience tells us that a commodity can be a good up to a point of satiation, beyond which it becomes a bad



Goods, bads, and neuters

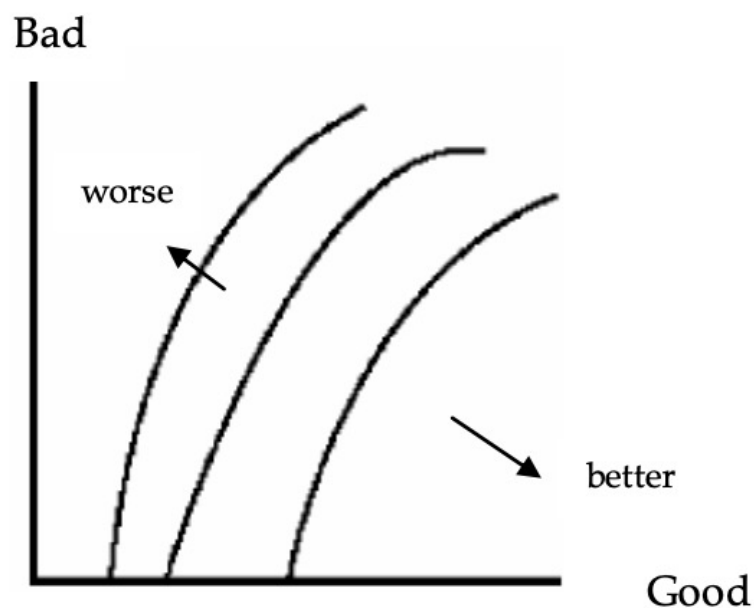
Portfolio preferences



Questions from last lecture

Lecture 3

- How would be a graph of a 'good' and a 'bad'?



Homework Lecture 3

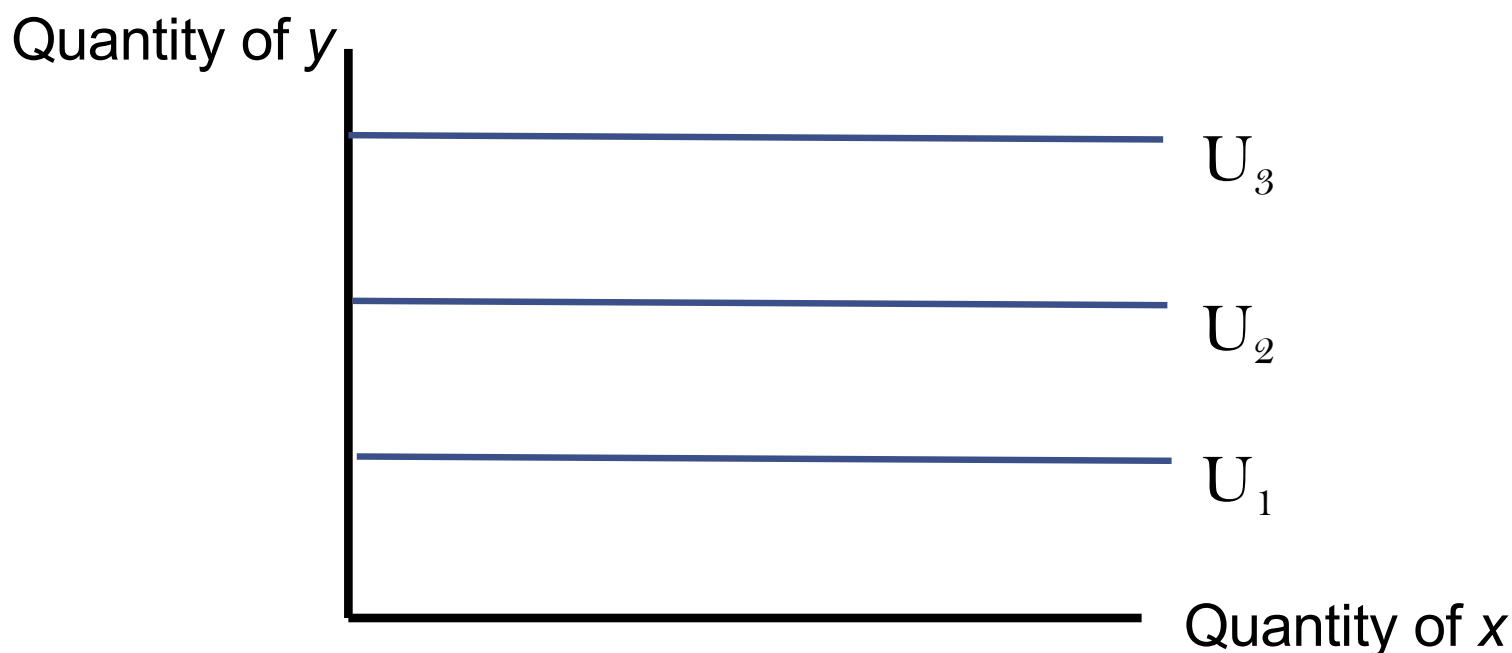
Draw a indifference curve map of a good (y) and a neuter commodity (x).

HINT: the consumer does not care about having more or less x

L3: Homework solution

Draw an indifference curve map of a **good** (y) and a **neuter commodity** (x).

HINT: the consumer does not care about having more or less x



Budget Constraint

Consumer Theory

3. Budget Constraint

Every consumer faces a **budget restriction** because of their limited income.

Video: <http://www.youtube.com/watch?v=-IzoID3Bx-0>

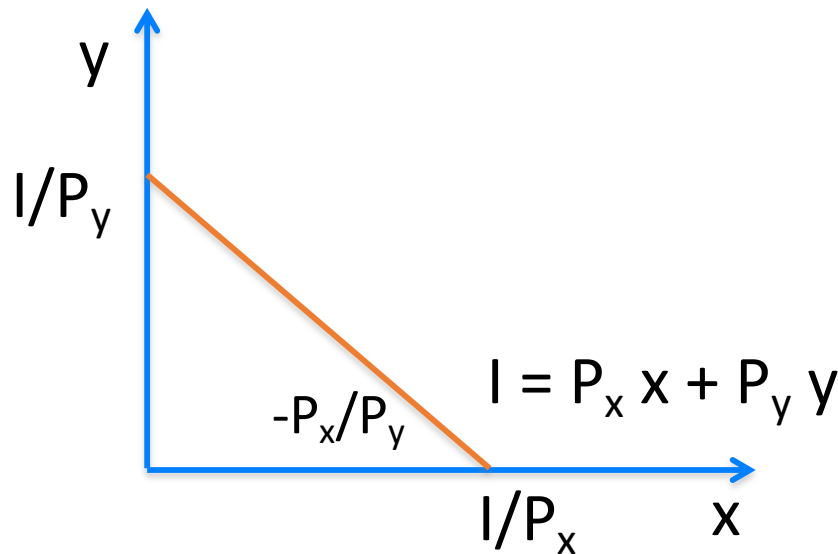
The budget constraint represents all the possible combinations where the quantity spent equals income (I).

Mathematically, the budget constraint is represented as:

$$P_x x + P_y y = I$$

3. Budget Constraint

$$P_x x + P_y y = I \rightarrow y = \frac{I}{P_y} - \frac{P_x}{P_y} x$$

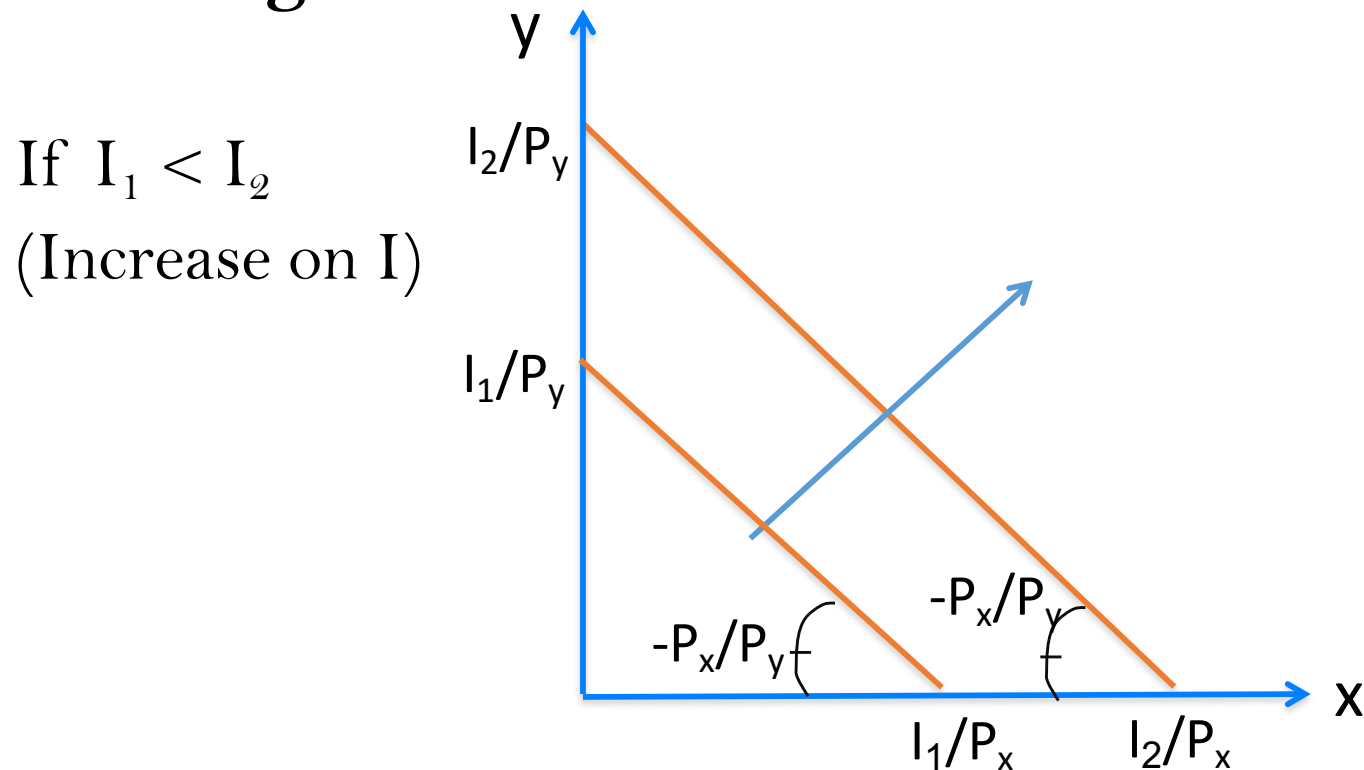


$-P_x/P_y$ (slope) shows how can y be traded for x in the market, keeping the money spent unchanged.

This ratio of prices is given by the market (it is something objective)

3. Budget Constraint

What happens to the budget constraint if there is a change in Income?

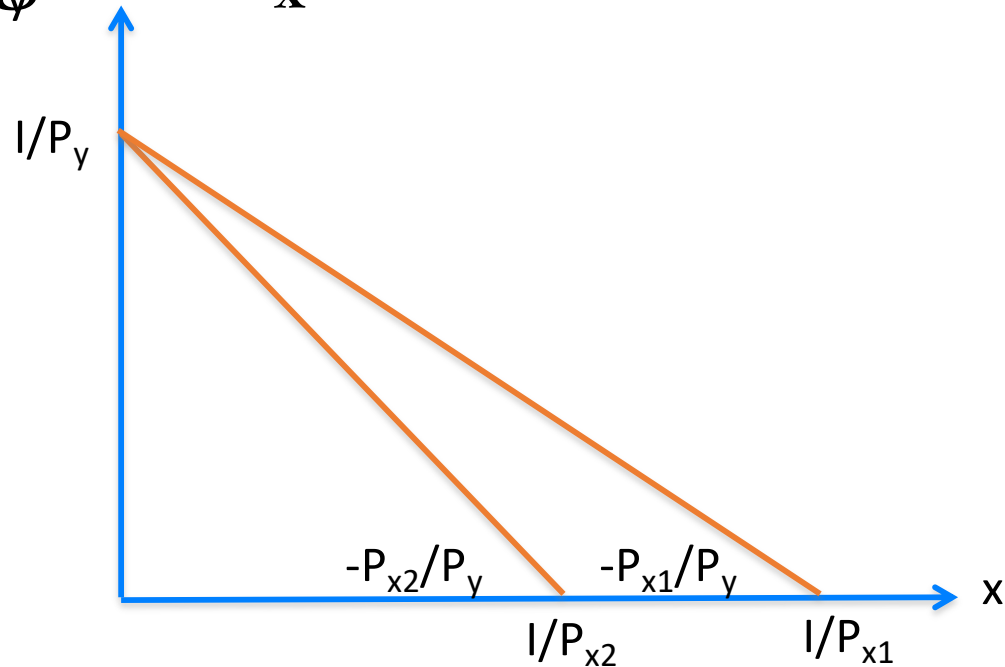


Budget constraint's **slope does not change.**

3. Budget Constraint

What happens to the budget constraint if there is a change in P_x ?

If $P_{x1} < P_{x2}$
(Increase on P_x)



Budget constraint's **slope does change**.

Activity 1

Purchasing power: Ability to generate utility by means of purchasing goods and services. Determined by income and prices.

1. What happens if P_x and P_y decrease to half of their values? What happens to purchasing power?
2. And if Income **also** decreases to half its original value? What happens to purchasing power?

Please answer using graphs to explain your answer.

Consumer's Choice

Consumer Theory

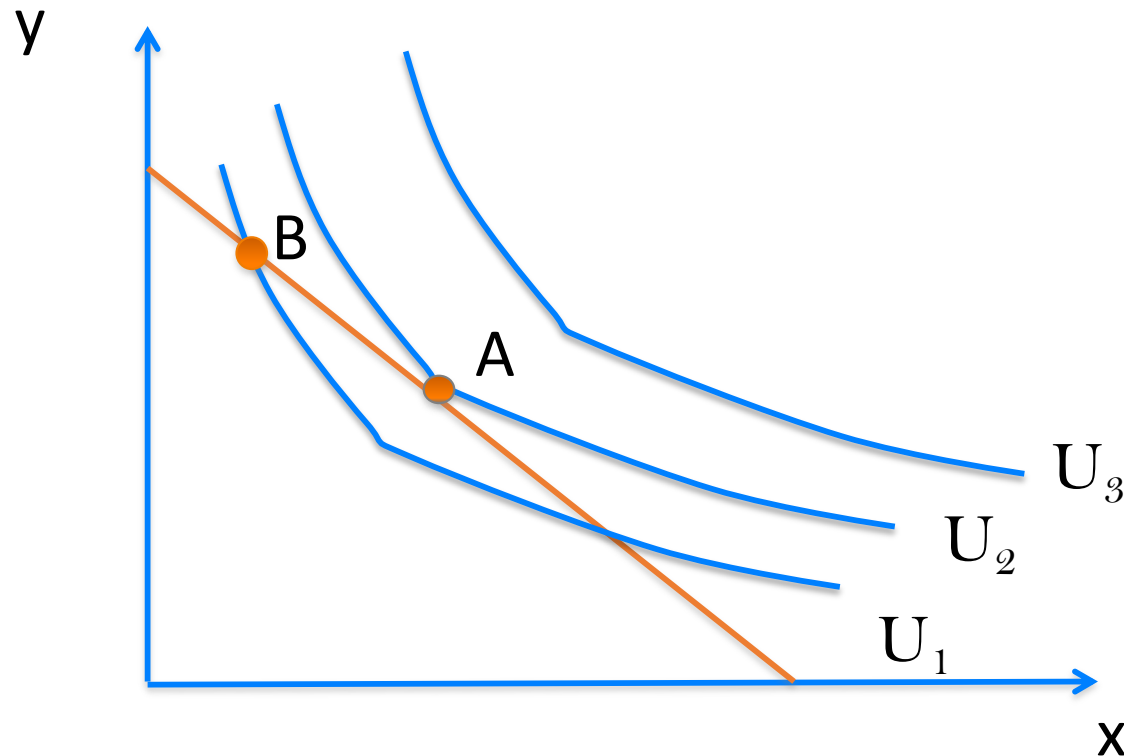
4. Consumer's Choice

We will assume that individuals are **rational** and will choose a consumption bundle that maximizes their satisfaction given their income

→ Looking for an **optimum consumption bundle**.

This optimum bundle should be **in the budget constraint**
→ individual must spend all her/his income.

4. Consumer's Choice



Point A should be the point of tangency between the budget constraint and the indifference curve (interior solution)

Meme 1 due next Friday

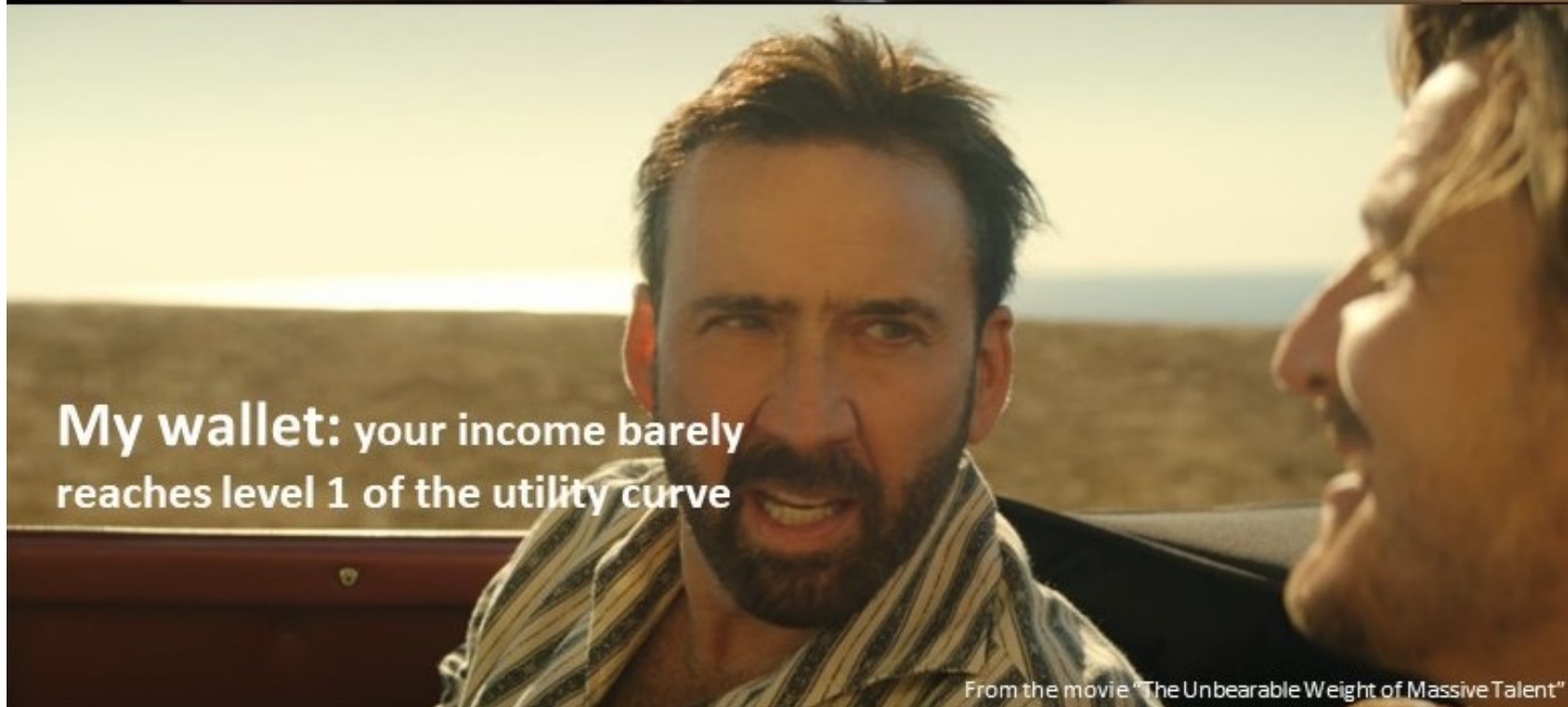


Background image from <https://www.ebaumsworld.com/pictures/21-popular-meme-templates-so-you-can-join-in-on-the-fun/85780323/?image=85780325>



Author: Diego Aguilera (2023)





Author: Paula Elicer (2023)

Meme 1

The thought of buying all the catnip to later resell.



The reality of facing budget restrictions



Author: Emilia Ureta (2023)

4. Consumer's Choice

Therefore, at point A:

Slope budget constraint = Slope indifference curve

OBJECTIVE

SUBJECTIVE

$$-\frac{P_x}{P_y} = -MRS = -\frac{MU_x}{MU_y}$$

$$\frac{MU_y}{P_y} = \frac{MU_x}{P_x}$$

Individual must adjust his consumption in order to achieve this equality

4. Consumer's Choice

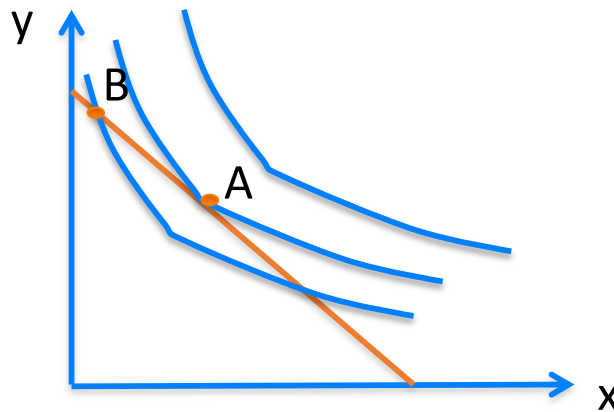
Value Paradox (diamonds and water):

Nothing is more useful than water: but it will purchase scarce anything; scarce anything can be had in exchange for it. A diamond, on the contrary, has scarce any use-value; but a very great quantity of other goods may frequently be had in exchange for it (Adam Smith)

$$\frac{MU_{Diamonds}}{P_{Diamonds}} = \frac{MU_{Water}}{P_{Water}}$$

4. Consumer's Choice

At point A: $-MRS = -P_x/P_y \rightarrow MU_x/P_x = MU_y/P_y$



At point B: $MRS > P_x/P_y \rightarrow MU_x/P_x > MU_y/P_y$

Attention: Corner solutions are the only ones where may happen that $MRS \neq P_x/P_y$

4. Consumer's Choice

Mathematical approach with n goods:

$$\textit{Max}_{\{x_1, x_2, \dots, x_n\}} U(x_1, x_2, \dots, x_n)$$

s.to :

$$I = P_1x_1 + P_2x_2 + \dots + P_nx_n$$

We can set up the Lagrangian expresion:

$$\mathcal{L} = U(x_1, x_2, \dots, x_n) + \lambda(I - P_1x_1 - P_2x_2 - \dots - P_nx_n)$$

4. Consumer's Choice

$$\mathcal{L} = U(x_1, x_2, \dots, x_n) + \lambda(I - P_1x_1 - P_2x_2 - \dots - P_nx_n)$$

First-order conditions (n+1 equations):

$$\frac{\partial \mathcal{L}}{\partial x_i} = \frac{\partial U}{\partial x_i} - \lambda P_i = 0, \text{ for } i=1 \text{ to } n$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = I - P_1x_1 - P_2x_2 - \dots - P_nx_n = 0$$

$$\begin{aligned} \text{First } n\text{-order conditions} &\rightarrow \frac{\frac{\partial U}{\partial x_i}}{P_i} = \lambda = \frac{\frac{\partial U}{\partial x_j}}{P_j} \\ &\rightarrow \text{MRS } (x_i \text{ for } x_j) = P_i/P_j \end{aligned}$$

4. Consumer's Choice

What does λ represent?

λ is a **shadow price** and for this optimization problem represents the **marginal utility of income**

$$\lambda = \frac{dU}{dI}$$

It represents the marginal utility generated when you relax the budget constraint by rising Income by \$1

4. Consumer's Choice

$$Utility = U(x_1, x_2, \dots, x_n)$$

$$dU = \frac{\partial U}{\partial x_1} dx_1 + \frac{\partial U}{\partial x_2} dx_2 + \dots + \frac{\partial U}{\partial x_n} dx_n$$

$$dU = MU_{x_1} dx_1 + MU_{x_2} dx_2 + \dots + MU_{x_n} dx_n$$

Remembering first n-order conditions:

$$dU = \lambda P_{x_1} dx_1 + \lambda P_{x_2} dx_2 + \dots + \lambda P_{x_n} dx_n$$

$$dU = \lambda (P_{x_1} dx_1 + P_{x_2} dx_2 + \dots + P_{x_n} dx_n)$$

$$dU = \lambda dI$$

$$\lambda = \frac{dU}{dI}$$

Example: Cobb-Douglas

An individual has a Cobb-Douglas utility function given by

$$U(x, y) = x^{\alpha} y^{\beta}, \text{ where } \alpha + \beta = 1$$

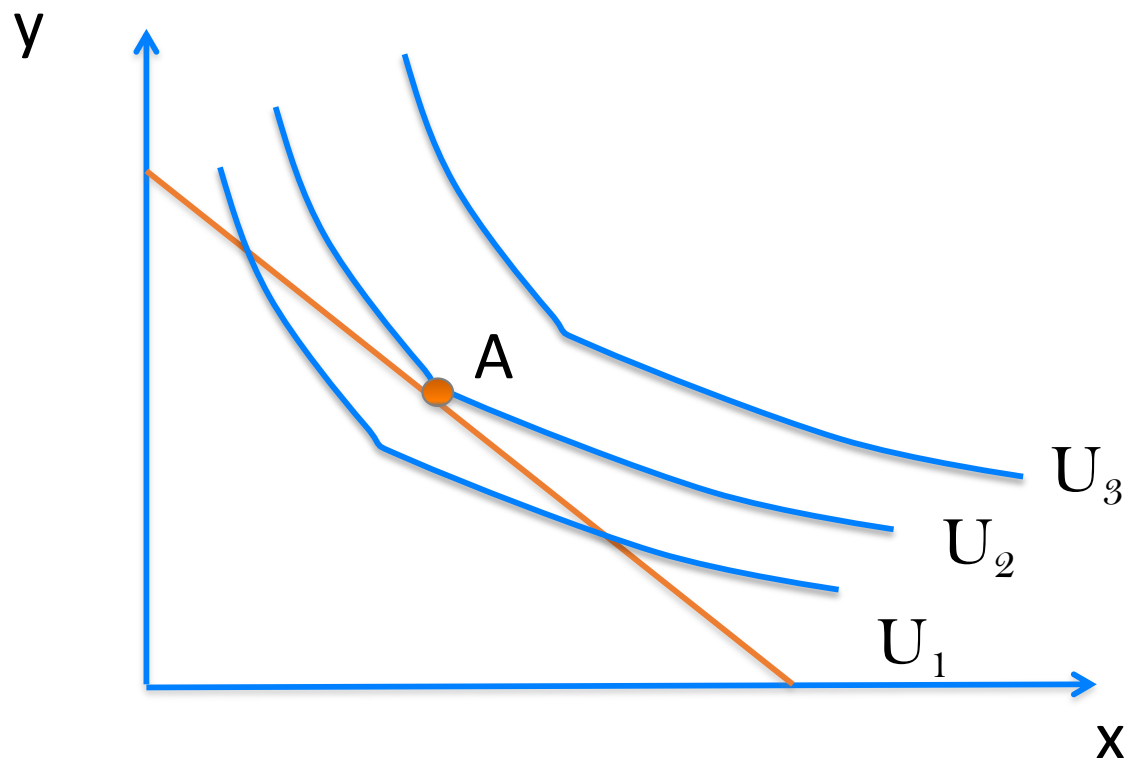
And has a budget constraint given by

$$P_x x + P_y y = I$$

Solve the utility maximizing values for x and y .

Example solution

Step 1: Before attempting to solve the problem mathematically, **check (graphically) if it is an interior solution or corner solution**



A Cobb-Douglas utility function given by $U(x, y) = x^\alpha y^\beta$, where $\alpha + \beta = 1$

$$\rightarrow MRS = \frac{P_x}{P_y}$$

Example solution

Step 2: Now that we know that we can look for the optimum consumption bundle mathematically, we use the utility maximization problem:

$$\text{Max}_{x,y} U(x, y) = x^\alpha y^\beta, \text{ where } \alpha + \beta = 1$$

Subject to:

$$P_x x + P_y y = I$$

Thus, the Lagrangian is

$$\mathcal{L}(x, y, \lambda) = x^\alpha y^\beta + \lambda(I - P_x x - P_y y)$$

Example solution

Step 3: Derive the first-order conditions

$$\mathcal{L}(x, y, \lambda) = x^\alpha y^\beta + \lambda(I - P_x x - P_y y)$$

$$\frac{\partial \mathcal{L}}{\partial x} = \alpha x^{\alpha-1} y^\beta - \lambda P_x = 0 \rightarrow \lambda = \frac{\alpha x^{\alpha-1} y^\beta}{P_x} \text{ (eq 1)}$$

$$\frac{\partial \mathcal{L}}{\partial y} = \beta x^\alpha y^{\beta-1} - \lambda P_y = 0 \rightarrow \lambda = \frac{\beta x^\alpha y^{\beta-1}}{P_y} \text{ (eq 2)}$$

Since it is the same $\lambda \rightarrow \frac{\alpha x^{\alpha-1} y^\beta}{P_x} = \frac{\beta x^\alpha y^{\beta-1}}{P_y} \rightarrow \frac{\alpha y}{\beta x} = \frac{P_x}{P_y} (*)$

Example solution

Step 3: Derive the first-order conditions
Another route for this step:

$$MRS = \frac{P_x}{P_y}$$
$$\frac{MU_x}{MU_y} = \frac{P_x}{P_y}$$

$$\frac{\alpha x^{\alpha-1} y^{\beta}}{\beta x^{\alpha} y^{\beta-1}} = \frac{P_x}{P_y}$$

$$\frac{\alpha y}{\beta x} = \frac{P_x}{P_y} (*)$$

Example solution

Step 3: Derive the first-order conditions

$$\mathcal{L}(x, y, \lambda) = x^\alpha y^\beta + \lambda(I - P_x x - P_y y)$$

The last first-order condition is:

$$\frac{\partial \mathcal{L}}{\partial \lambda} = I - P_x x - P_y y = 0$$

Replacing (*) in this condition yields

$$I - P_x x - P_y \frac{P_x \beta x}{\alpha P_y} = 0$$

$$P_x x \left(1 + \frac{\beta}{\alpha}\right) = I$$

$$x^* = I \frac{\alpha}{\alpha + \beta} \frac{1}{P_x} = \alpha \frac{I}{P_x}, y^* = \beta \frac{I}{P_y}$$

Example: Cobb-Douglas

An individual has a Cobb-Douglas utility function given by $U(x, y) = x^\alpha y^\beta$, where $\alpha + \beta = 1$

And has a budget constraint given by

$$P_x x + P_y y = I$$

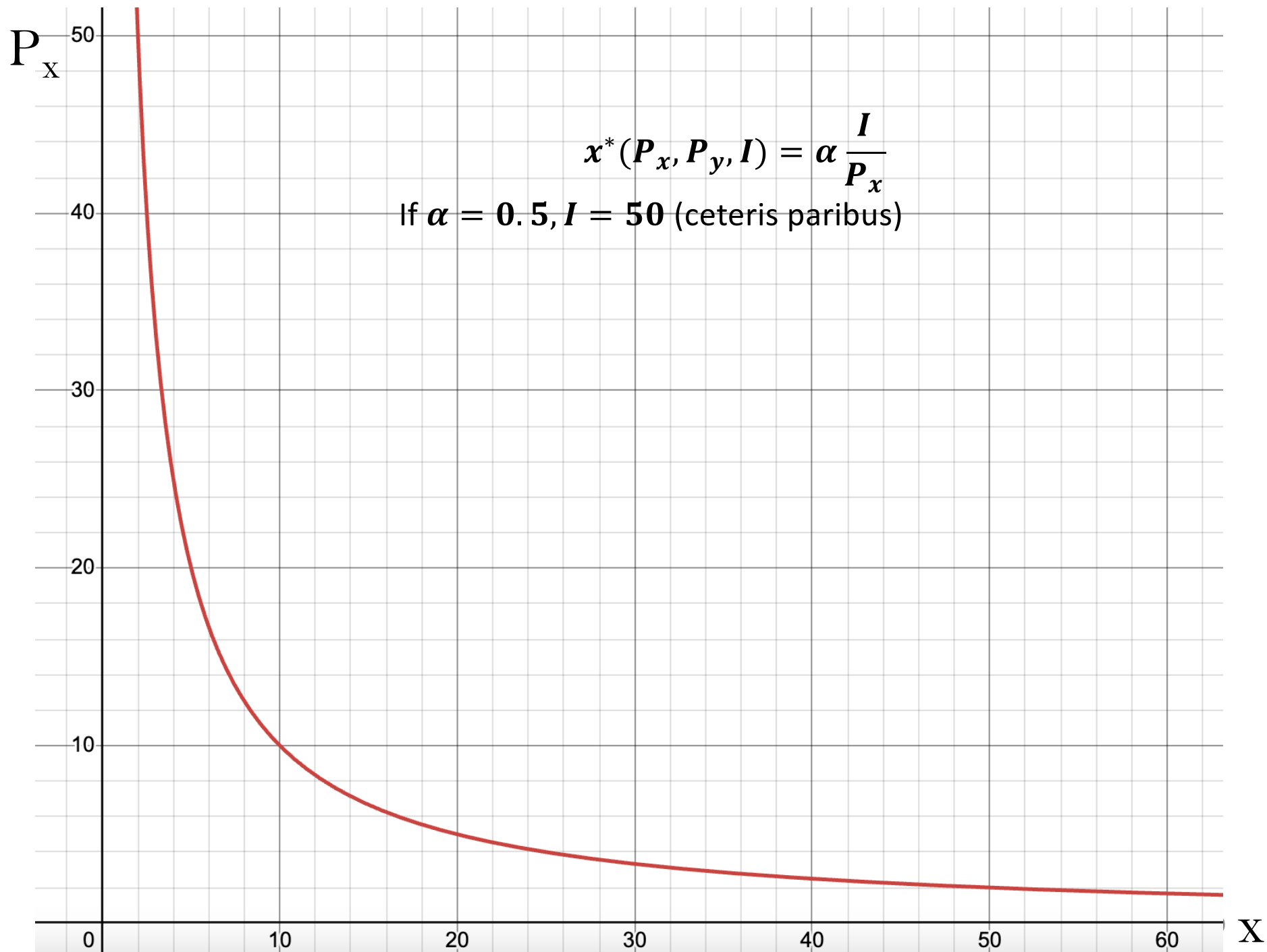
Solve the utility maximizing values for x and y .

The optimal values are:

$$x^*(P_x, P_y, I) = I \frac{\alpha}{\alpha + \beta} \frac{1}{P_x} = \alpha \frac{I}{P_x}$$

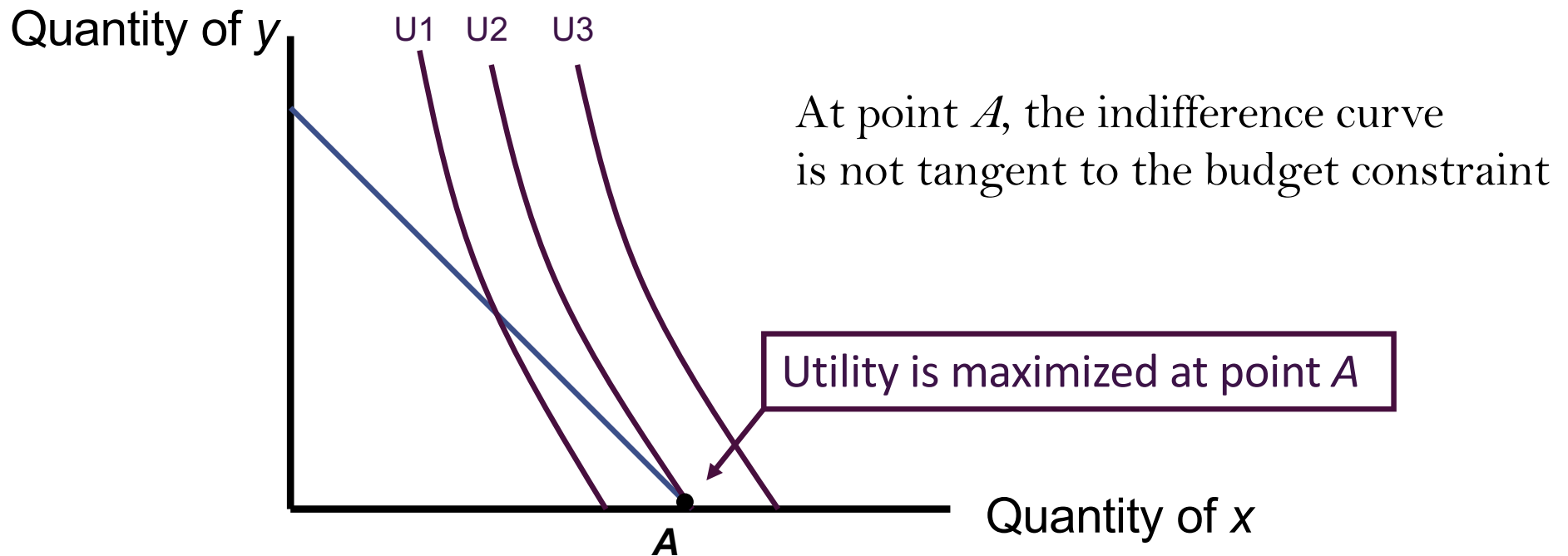
$$y^*(P_x, P_y, I) = \beta \frac{I}{P_y}$$

Demand
of x and
 y , given
 P_x, P_y, I



4. Consumer's Choice

Corner solutions are the only ones where may happen that $MRS \neq P_x/P_y$



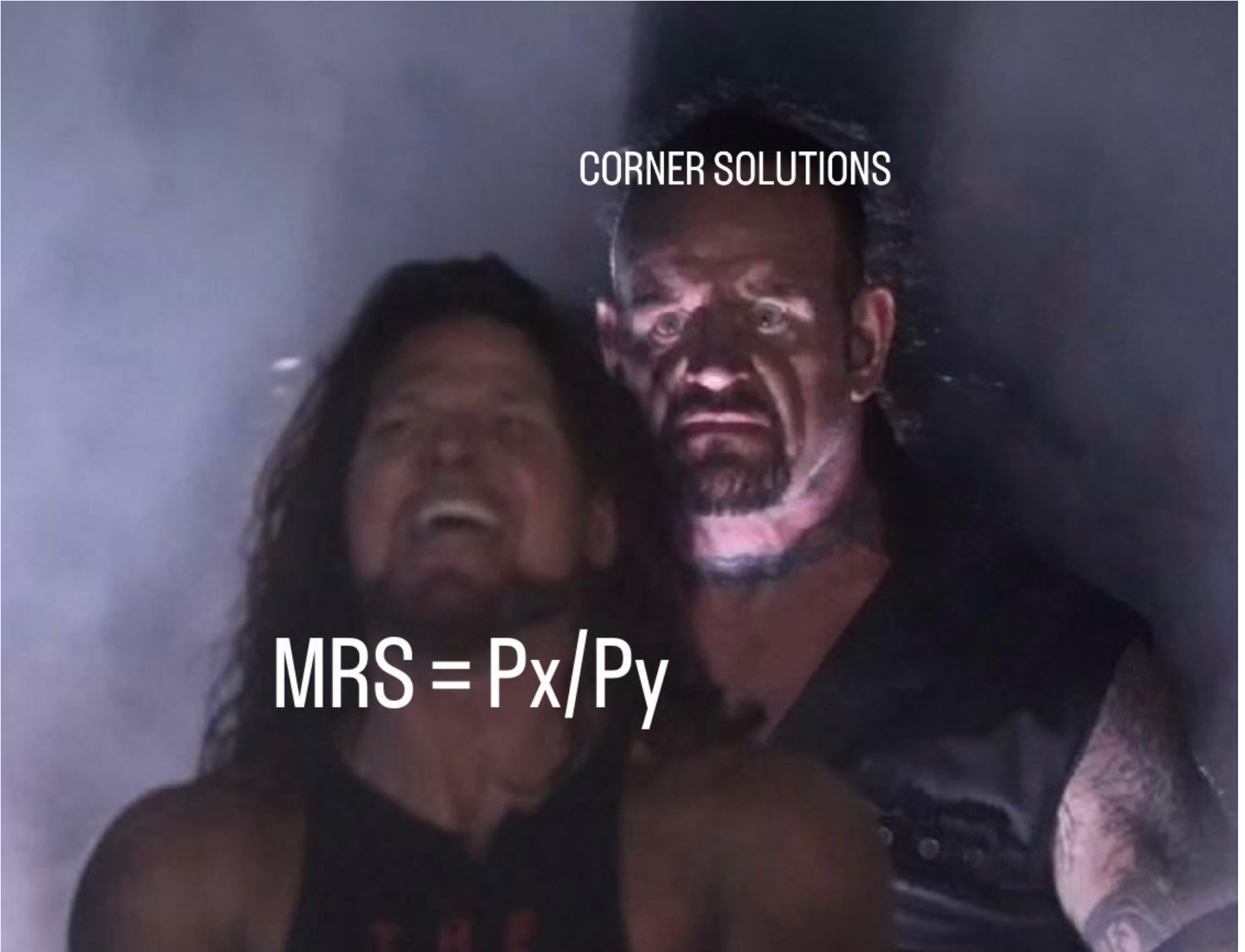
you told me in the optimum point of
your consumption bundle the MRS is
different of P_x/P_y

in corner solutions



Recovered from: <https://www.telesurtv.net/news/origen-meme-viral-gato-blanco-redes-sociales--20190905-0020.html>

Credits to: Patricia Valenzuela



CORNER SOLUTIONS

$$\text{MRS} = P_x/P_y$$

4. Consumer's Choice

What happens with corner solutions?

First-order conditions must be modified:

$$\partial \mathcal{L} / \partial x_i = \partial U / \partial x_i - \lambda p_i \leq 0 \quad (i = 1, \dots, n)$$

(Kuhn-Tucker conditions)

If $\partial \mathcal{L} / \partial x_i = \partial U / \partial x_i - \lambda p_i < 0$, then $x_i = 0$

$$p_i > \frac{\partial U / \partial x_i}{\lambda} = \frac{MU_{xi}}{\lambda}$$

Kuhn-Tucker conditions (KKT)

$$\begin{array}{l} \text{Max } f(x) \\ \text{Subject to: } g(x) \leq c \end{array}$$

Define the Lagrangian:

$$\mathcal{L}(x) = f(x) - \lambda(g(x) - c)$$

First-order conditions:

$$\frac{\partial \mathcal{L}}{\partial x_i} = \frac{\partial f(x)}{\partial x_i} - \lambda \frac{\partial g(x)}{\partial x_i} = 0$$

$$(for\ i = 0, 1, \dots, n)$$

$$\lambda \geq 0, \quad \lambda \cdot [g(x) - c] = 0$$

$$g(x) \leq c$$

$$\begin{array}{l} \text{Max } U(x) \\ \text{Subject to: } \sum_{i=1}^n P_i x_i \leq I \end{array}$$

$$\mathcal{L}(x) = U(x) - \lambda \left(\sum_{i=1}^n P_i x_i - I \right)$$

$$\frac{\partial \mathcal{L}}{\partial x_i} = \frac{\partial U(x)}{\partial x_i} - \lambda P_i = 0$$

$$(for\ i = 0, 1, \dots, n)$$

$$\lambda \geq 0, \lambda \cdot \left[\sum_{i=1}^n P_i x_i - I \right] = 0$$

$$\sum_{i=1}^n P_i x_i \leq I$$

One-minute paper - Canvas

Write down **2-3 concepts** that you learnt today. Please write down the concept and what you learnt about it, in your own words.

If you have any **questions** about we discussed today, write it down too. We will go through them at the beginning of next class.

Finally, if there is a word you didn't understand, write it down so it can be part of the **glossary**.

Activity 2

If an individual has a budget constraint given by:

$$P_x x + P_y y = I$$

Find his/hers optimum consumption bundle if goods x and y are:

a) Perfect Substitutes: $U(x, y) = \alpha x + \beta y$

Assume that $\alpha/\beta < P_x/P_y$

b) Perfect Complements: $U(x, y) = \text{Min}\{x, y\}$

HINT: work graphically